

PRODUCT OVERVIEW

controllerX360 is a vendor-agnostic control-loop performance monitoring application. It reads PV, SP, OP and MODE for every controller from the unit's own historian and passes each one through a cascading diagnostic tree — data quality, service factor, saturation, stiction, oscillation, sluggish tuning — so that a mechanical or operational fault is isolated *before* any tuning advice is given. No verdict rests on a single test: stiction, the hardest call to make, is confirmed only when four independent detectors agree. The output is not another list of alarms but a ranked, evidence-backed work list — which loops, what is wrong, and who owns the fix. Every threshold is normalised to the configured PV range, so a 180 t/h flow loop and a 1 mol% purity analyser are held to the same standard.

01 Unit Configuration

Shown: the five-step configuration wizard, the controller register and the active rule profile.

01

Know your unit

Historian, window, scan rate

Configured

02

Controller register

PV / SP / OP / MODE tags

50 of 50

03

Diagnostic agents

Engage per stage and method

7 of 7

04

Rules & thresholds

Gates, modes, exclusions

Validated

05

Review & run

Readiness, then execute

Ready

TAG	SERVICE	TYPE	UOM	RANGE	CONSEQUENCE	USE
15FC306	Ethylene rundown flow	FC	Kg/Hr	0 - 42 000	Critical	✓
15TC401	C2 splitter tray-42 temp	TC	degC	-40 - 10	Critical	✓
15PC602	Deethaniser pressure	PC	Kg/cm²	0 - 30	High	✓

ACTIVE RULE PROFILE

GATE	VALUE	BASIS
Service factor floor	85 %	AUTO + CAS + RCAS
Saturation dwell	30 %	samples at a limit
Stiction consensus	40 / 70	watch / confirmed
Hägglund regularity	0.60	zero-crossing spread
Harris index floor	0.30	vs minimum variance

Register — every controller with its tag map, range and consequence class. **Agents** — engage or disengage any stage, down to the individual stiction method. **Historian** — plug and play with PI, PHD, Exaquantum or IP.21, or an Excel upload. **Exclusions** — suppress a detection per loop where the behaviour is known and accepted.

02 Diagnosis and Prioritisation

Shown: a 50-loop ethylene fractionator benchmark — the verdict set, and where each gate catches its loops.

THE SIX PROBLEM TYPES — AND ONE CLEAN VERDICT

Low service factor 14%

Abandoned by the operators — nothing below it can be trusted.

Saturation 8%

Output pinned at a limit — a capacity fault, never a tuning one.

Stiction 8%

Stem friction, four methods agreeing — the valve needs work.

Oscillation — aggressive 12%

PV and OP both cycle — the controller drives it.

Oscillation — external 8%

PV cycles, OP barely moves — the source is upstream.

Sluggish 10%

Stable but far from minimum variance — recovery is slow.

Healthy 40%

Passed every gate — PID parameters match the process dynamics; kept on the benchmark so drift is caught early.

HOW IT WORKS — THE DIAGNOSTIC CASCADE

GATE	FC	TC	LC	PC	AC	CAUGHT	LEFT
1 Service factor	1	3	-	3	-	7	43
2 Saturation	2	2	-	-	-	4	39
3 Stiction	1	1	1	1	-	4	35
4 Oscillation	5	-	1	1	3	10	25
5 Sluggish	2	1	2	-	-	5	20
Healthy	7	3	5	3	2	20	40%

Controllers caught 0 → 5+ FC flow · TC temp · LC level · PC press · AC analyser

Why the order matters. Filtering operational state first keeps the classifiers from telling you to “reduce the gain” on a loop that is in MAN, pinned at a limit or stuck.

03 AI-Powered, Industry-Standard Methods

Four independent stiction detectors that must agree before the call is made, plus two benchmarks that grade the tuning.

Heuristic

OP ramps while PV stays still, then jumps — as an engineer reads it.

Horch cross-correlation

Time-delay asymmetry: separates stiction from aggressive tuning.

Yamashita phase plot

PV against OP: a line is a free valve, a parallelogram is stiction.

Choudhury bicoherence

Higher-order frequency statistics, robust to Gaussian noise.

Harris index

Grades loop variance against the minimum-variance benchmark.

Autocorrelation

Error autocorrelation and zero-crossing rate confirm a real cycle.

APPLIED ACROSS INDUSTRIES

Petrochemicals

Refining

Power and utilities

Metals and mining

Specialty chemicals

Pharma and life sciences

Fertilizers, agrochem

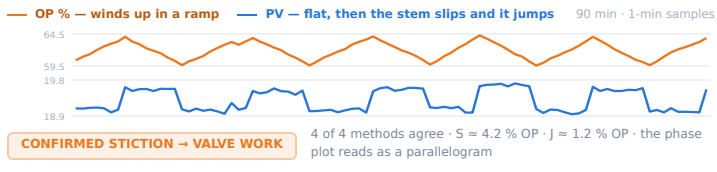
Pulp, paper, cement

Any plant with time-stamped PV / SP / OP / MODE data. The same gates and consensus logic transfer site to site — only the controller register changes.

04 Operator Experience

Controller-level detail: the evidence behind a verdict, and the ranked work list it produces.

VALVE STICTION — DEETHANISER COLUMN PRESSURE



WHAT THE TEAM RECEIVES — RANKED WORK LIST

#	CONTROLLER	DIAGNOSIS	HEALTH	ACTION & OWNER
1	15TC401	Saturation	12	Process intervention — <i>do not re-tune</i>
2	15FC318	Stiction	28	Valve shop — repack / positioner
3	15PC602	Stiction	29	Valve shop — repack / positioner
4	15FC310	Loop in MAN	37	Operations — re-qualify the loop
5	15FC301	Aggressive tuning	31	Control engineering — reduce gain

Ranked by consequence x severity — each row carries its evidence, the metric that fired and the owner of the fix.

05 Business Impact

What a correctly ordered diagnosis is worth to the operation.

100 % loop coverage

Every controller, not a sample.

4 of 4 method consensus

Four detectors must agree.

Zero false re-tunes

Faults isolated first.

6 + 1 verdicts

Six problem types plus healthy.

24x7 benchmarking

Continuous or one-off.

One list every discipline

One ranking, every discipline.

06 Platform

Runs inside your network; no data leaves the plant boundary.

Historian connectivity

Plug and play with major historians.

Live or on demand

Continuous or monthly runs.

Reporting

Downloadable interactive reports.

Gen AI assistant

Answers loop questions in context.

Build your own dashboard

Engineers build their own views.

Quick setup

Running within two weeks.